A LETTERS JOURNAL EXPLORING THE FRONTIERS OF PHYSICS

LETTER

Unification of Bell, Leggett-Garg and Kochen-Specker inequalities: Hybrid spatio-temporal inequalities

To cite this article: Siddhartha Das *et al* 2013 *EPL* **104** 60006

View the article online for updates and enhancements.

You may also like

- Probing inequivalent forms of Leggett–Garg inequality in subatomic systems
Javid Naikoo, Swati Kumari, Subhashish Banerjee et al.

- Violation of the Leggett-Garg Inequality in neutrino oscillations
T. E. Weiss, J. A. Formaggio, D. I. Kaiser et al.

- From micro- to macrorealism: addressing experimental clumsiness with semi-weak measurements
Devashish Pandey, Xavier Oriols and Guillermo Albareda

This content was downloaded from IP address 78.153.180.226 on 17/03/2026 at 11:19

A Letters Journal Exploring the Frontiers of Physics

December 2013

EPL, **104** (2013) 60006
doi: 10.1209/0295-5075/104/60006

www.epljournal.org

Unification of Bell, Leggett-Garg and Kochen-Specker inequalities: Hybrid spatio-temporal inequalities

Siddhartha Das[1(a)], S. Aravinda[2(b)], R. Srikanth[2,3(c)], Dipankar Home[4(d)]

[1] *Indian Institute of Science Education and Research - Pune-411008, India*
[2] *Poornaprajna Institute of Scientific Research - Sadashivnagar, Bengaluru-560080, India*
[3] *Raman Research Institute - Sadashivnagar, Bengaluru-560060, India*
[4] *CAPSS, Bose Institute - Kolkata-700 091, India*

received 26 August 2013; accepted in final form 2 January 2014
published online 24 January 2014

PACS 03.65.Ta – Foundations of quantum mechanics; measurement theory
PACS 03.65.Ud – Entanglement and quantum nonlocality
PACS 42.50.Xa – Optical tests of quantum theory

Abstract – The Bell-type (spatial), Kochen-Specker (contextuality) or Leggett-Garg (temporal) inequalities are based on classically plausible but otherwise quite distinct assumptions. For any of these inequalities, satisfaction is equivalent to a joint probability distribution for all observables in the experiment. This implies a joint distribution for all pairs of observables, and is indifferent to whether or not they commute in the theory. This indifference underpins a unification of the above inequalities into a general framework of correlation inequalities. When the physical scenario is such that the correlated pairs are all compatible, the resulting correlation is nonsignaling, which may be local or multi-particle, corresponding to contextuality or Bell-type inequalities. If the pairs are incompatible, the resulting correlation corresponds to Leggett-Garg (LG) inequalities. That quantum mechanics (QM) violates all these inequalities suggests a close connection between the local, spatial and temporal properties of the theory. As a concrete manifestation of the unification, we extend the method due to Roy and Singh (*J. Phys. A*, **11** (1978) L167) to derive and study a new class of hybrid spatio-temporal inequalities, where the correlated pairs in the experiment are both compatible or incompatible. The implications for cryptography and monogamy inequalities of the unification are briefly touched upon.

Copyright © EPLA, 2013

Introduction. – Three types of statistical *correlation inequalities* may be discerned that probe the nonclassical aspects of quantum mechanics (QM), namely: Bell-type or spatial inequalities [1,2], which test the assumption of local realism [3], Kochen-Specker [4] or contextuality inequalities [5–7] that test noncontextual realism, and finally Leggett-Garg or temporal inequality [8–12], that test macro-realism or noninvasive realism. Classical theory presumes *realism*, *i.e.*, observations merely reveal a pre-existing value of a measured property. The quantum-mechanical violation of these correlation inequalities implies that realism must sometimes be given up, unless classically counter-intuitive modes of influence are posited (for example, an explanation of the violation of Bell-type inequality would require signaling over space-like-separated events).

In the Bell-type and contextuality inequalities, the experimental *correlata* (the experimentally correlated pair of observables) are mutually compatible (or commuting, in QM), whereas in the LG inequality, sometimes considered as the temporal analog of a Bell-type inequality, the correlata are incompatible, necessitating the measurements to be sequential. Further, we may regard Bell-type inequalities as a special case of contextuality inequalities, wherein context is provided through spatial separation. All the above inequalities have been tested, modulo certain arcane loopholes, in laboratories worldwide: for Bell-type inequalites [13–16], for contextuality inequalities [17,18] and LG inequalities in both macroscopic systems [19,20] and microscopic systems [21–24].

The problem of unifying these inequalities into a single framework was first considered by Markiewicz *et al.* [25],

(a) E-mail: sdas@students.iiserpune.ac.in
(b) E-mail: aru@poornaprajna.org
(c) E-mail: srik@poornaprajna.org
(d) E-mail: dhome@jcbose.ac.in

who showed that the existence of a joint probability distribution is the general assumption underlying all three types of inequalities. As one consequence, they demonstrated how one type of inequality could be transformed into another. In this work, we examine another consequence of the unification, namely that of hybrid inequalities, which mix correlations of different types. Here we make use of the interesting method due to Roy and Singh [26,27], who were the first to derive from the local-realist condition testable inequalities *different* from the Bell-CHSH–type inequalities. We adapt the Roy-Singh (RS) method of deriving spatial inequalities to derive contextual, temporal and finally hybrid (spatio-temporal) inequalities. The new inequalities we obtain are no harder to implement experimentally than the other inequalities.

The remaining part of this paper is divided as follows. In the following section, we introduce the RS method for deriving Bell-type inequalities. Later, we point out that the existence of a joint distribution underlies these inequalities, essentially revisiting a theorem due to Fine [28] in the context of the RS inequalities. That this observation also extends to the case of contextuality and LG-type inequalities is shown subsequently. As a demonstration of the generality of this result, we next derive contextuality and LG-type inequalities via the RS method, followed by a novel class of hybrid spatio-temporal inequalities. After a discussion on monogamy inequalities, we present conclusions in the last section, touching on some cryptographic ramifications of our work.

RS inequalities. – The RS method provides an elegant method to derive local-realist inequalities [26]. Suppose two qubits in an entangled state are intercepted by two measuring devices geographically separated from each other. The first device randomly measures property either $X_1, X_2, \cdots$ on particle 1, and the other device either property $Y_1, Y_2, \cdots$ on particle 2. Experimentally, one measures bi-partite correlations of the type $P(x_j, y_k)$, where $x_j = \pm 1$ and $y_k = \pm 1$ are measurement outcomes of measuring X_j and Y_k respectively. The assumption of local realism (LR) entails that for each such pair of these variables, there is a deterministic hidden variable (DHV) theory whereby

$$\langle X_j Y_k \rangle = \int_\lambda \mathrm{d}\lambda \rho(\lambda) X_j(\lambda) Y_k(\lambda), \tag{1}$$

where λ is a "complete" or "dispersion-free" specification of the state described by the underlying probability distribution $\rho(\lambda)$. Roy and Singh [26] consider quantities of the form

$$\begin{aligned}&\left(X_1^{(1)} \pm X_2^{(1)} \pm \cdots X_{m_1}^{(1)} + Y_1^{(1)} \pm Y_2^{(1)} \pm \cdots \pm Y_{n_1}^{(1)}\right)^2 \\ &+ \left(X_1^{(2)} \pm X_2^{(2)} \pm \cdots X_{m_2}^{(2)} + Y_1^{(2)} \pm Y_2^{(2)} \pm \cdots \pm Y_{n_2}^{(2)}\right)^2 \\ &+ \cdots + \left(X_1^{(q)} \pm X_2^{(q)} \pm \cdots X_{m_q}^{(q)} \right. \\ &\left. + Y_1^{(q)} \pm Y_2^{(q)} \pm \cdots \pm Y_{n_q}^{(q)}\right)^2 \geq q,\end{aligned} \tag{2}$$

where $m_j + n_j =$ odd, and the self-correlation terms, *i.e.*, correlations between the same particle, are so arranged as to cancel out. Here $X_k^{(j)} \in \{X_1, X_2, \cdots, X_n\}$ and $Y_k^{(j)} \in \{Y_1, Y_2, \cdots, Y_m\}$, where m, n are positive integers and $X_j, Y_k \in \{\pm 1\}$. As a particular example, we consider

$$(X_1 - Y_1 - Y_2)^2 + (X_2 - Y_1 + Y_2)^2 \geq 2. \tag{3}$$

Expanding the lhs of (3), and using the DHV assumption eq. (1), we obtain

$$\langle \mathcal{B} \rangle \equiv \langle X_1 Y_1 \rangle + \langle X_1 Y_2 \rangle + \langle X_2 Y_1 \rangle - \langle X_2 Y_2 \rangle \leq 2, \tag{4}$$

which just is the CHSH inequality [1]. By varying the odd-termed expressions in eq. (3), in such a way that the same-particle correlations (self-correlations) cancel out, Roy and Singh obtain other Bell-type inequalities [26].

Joint realism. – In the above derivation of a Bell-type inequality, the following point is worth noting: that in the step represented by eq. (2), one presumes *joint realism* (JR) of all variables in the experiment, *i.e.*, that there exists a joint probability distribution (JD) $P(x_1, x_2, \cdots, x_n, y_1, y_2, \cdots, y_m)$ of all variables in the experiment. This is equivalent to the DHV considered in eq. (1) [28]. To see this, we note that the existence of a DHV defines a JD

$$\begin{aligned}&P(x_1, \cdots, x_n, y_1, \cdots, y_m) = \\ &\int_\lambda \rho(\lambda) \mathrm{d}\lambda \tilde{X}_1(x_1, \lambda) \tilde{X}_2(x_2, \lambda) \cdots \tilde{X}_n(x_n, \lambda) \\ &\times \tilde{Y}_1(y_1, \lambda) \tilde{Y}_2(y_2, \lambda) \cdots \tilde{Y}_m(y_m, \lambda),\end{aligned} \tag{5}$$

where $\tilde{X}_j(x_j, \lambda)$ is an indicator function that takes the value 1 if $X_j = x_j$ given λ, else 0, similarly for $\tilde{Y}_k(y_k, \lambda)$. This JD corresponds to the two-particle correlations of eq. (1) since

$$\begin{aligned}\langle X_j Y_k \rangle &\equiv \sum_{x_1 \cdots, y_1, \cdots = \pm 1} P(x_1, x_2, \cdots, x_n, y_1, y_2, \cdots, y_m) x_j y_k \\ &= \sum_{x_j, y_k} P(x_j, y_k) x_j y_k \\ &= \int_\lambda \rho(\lambda) \sum_{x_j, y_j} \tilde{X}_j(x_j, \lambda) \tilde{Y}_k(y_k, \lambda) x_j y_k\end{aligned} \tag{6}$$

from which the rhs of eq. (1) follows since by definition $\sum_{x_j} \tilde{X}_j(x_j, \lambda) x_j = X_j(\lambda)$ and $\sum_{y_k} \tilde{Y}_k(y_k, \lambda) y_k = Y_k(\lambda)$. That JR implies that Bell-type inequalities hold follows from the previous section. The converse of these two arguments can be given along the lines adopted in ref. [28], thereby establishing the equivalence of the existence of JD, of DHV and the satisfaction of correlation inequalities.

Unification. – A key observation is that the satisfication of a correlation inequality implies that consistent JDs exist for all commuting *and noncommuting* pairs of observables. The proof follows from the fact that the required

JDs, say $P(x_1, y_2)$, can be obtained by tracing out all the other variables from $P(x_1, \cdots, x_n, y_1, \cdots, y_m)$. Suppose $S = A \cup B$, where $A \equiv \{X_j\}$ and $B \equiv \{Y_k\}$ are two groups of properties, and the experimental outcome is a correlation of the form $P(x_j, y_k)$. The *instantiation* of the grouping does not matter: in other words, A and B may refer to two geographically separated particles, or to two different times on the same particle, and so forth.

From the perspective of an experiment, there are two cases to consider:

Compatible correlata. That all elements within A are mutually incompatible, as are all elements within B, whereas every element X_j is compatible with every element Y_k. This compatibility means that the correlation $P(x_j, y_k)$ will be nonsignaling, since measurements in A reveal no information about operations in B and vice versa. A correlation inequality obtained here may thus be referred to as *nonsignaling*. measured cross-correlations are between compatible observables, and the cross-correlated observables have pairwise JD, the violation of a correlation inequality in this case arises from local incompatibility within A and/or within B. This explains the close connection between noncommutativity [29] or complementarity [30] or Heisenberg uncertainty [31] on the one hand and nonlocality on the other.

Incompatible correlata. That elements between A and B, in addition to having intra-group incompatibility, also have *inter-group* incompatibility. When A is a set of measurements on a particle at time t_A, and B is a set of measurements on the same particle at time $t_B > t_A$. Correlations are obtained by *sequential* measurements on account of the incompatability. The correlation $P(x_j, y_k)$ will be signaling, since the outcome of Y_k will depend on which X_j preceded it [32]. Here we are led to *signaling* correlation inequalities. The term signal is used in the formal sense that the no-signaling condition is violated, and does not refer to signal between spatially seperated parties. By virtue of relativistic causality, signaling correlations cannot pertain to spacelike-separated measurements on geographically separated particles but they can describe sequential measurements on the same particle.

Clearly, Bell-type and contextuality inequalities are of the nonsignaling kind, while the temporal inequalities are necessarily of the signaling kind. Our observation at the beginning of this Section implies that a correlation inequality is indifferent as to whether the correlation is signaling or otherwise: in both cases, violation of an inequality means that a HDV or equivalent JD description is not possible. This indifference means that we can regard the spatial, temporal and contextual inequalities as instantiations of the same unified mathematical object, which is a HDV or a JD.

The physical significance of this object depends on the physical scenario at hand. Let A and B correspond to geographically separate observers. Assuming JR, a violation of the inequality, implies a microscopic spacelike signal in the sense of B's outcome depending on A's setting or vice versa. Thus satisfaction of the inequality is tantamount to the assumption of LR. Analogously, if A and B correspond to the same particle or system, then JD is equivalent to the assumption of noncontextual realism (NCR). On the other hand, in the case of a signaling correlation inequality, A and B necessarily pertain to the same particle. In this case JD requires that subsequent measurements were not disturbed by the earlier ones, and thus tantamount to the assumption of noninvasive realism (NIR) or macro-realism.

There are profound ramifications of this unification. One is that the method to derive any one type of inequality can be translated in straightforward fashion to yield another type of inequality [25]. Accordingly, we adapt the RS method to derive contextuality and LG-type inequalities. Second is that, in all three types of inequalties , the *classical bound*, which is the rhs (having value 2 in eq. (4)) is indifferent to whether a spatial, temporal or contextual scenario is instantiated. The Tsirelson bound is also identical for the considered cases involving qubits, though in general this bound can be different [25]. Another ramification is that QM violates these inequalities in all three instantiations, meaning that its nonclassical structure exists under spatial, temporal and contextual instantiations, showing a deep connection between these aspects. One can in principle imagine toy theories where this three-fold nonclassicality need not hold true, for example, classical theory, genaralized local theory [33] equipped with a temporal PR box for sequential measurements.

As a final ramification of the unification, we demonstrate below using the RS method *hybrid inequalities* in which some of the correlata are compatible and some are not. Such an inequality will be a simultaneous test of, say, NIR and LR, which is noninvasive local realism (NILR). One can similarly instatiate JR to obtain an inequality that tests NCR and LR together, or NIR and LR together or all three together. A spatio-temporal hybrid inequality is considered in detail later below.

Contextuality and temporal inequalities from the RS method. – The generality of the observation that JR underlies all the three types of inequalities is seen by adapting the RS method to derive spatial and contextuality inequalities.

As a prelude to deriving the hybrid inequality via the RS method, we discuss how one can obtain contextuality and LG-type inequalities. It is clear that the RS procedure already yields state-dependent contextuality inequalities when the labels X and Y of the Bell-type inequalities are interpreted as defining pairwise compatible properties of the same particle.

The contextuality inequality studied by ref. [5] can be obtained starting from

$$(X_1+X_2+X_3)^2+(X_3+X_4+X_5)^2+(X_1-X_3+X_5)^2 \geq 3, \tag{7}$$

which yields

$$X_1X_2 + X_2X_3 + X_3X_4 + X_4X_5 + X_5X_1 \geq -3, \quad (8)$$

where the consecutive subscript indices refer to commuting observables of the same particle. This can be generalized in a similar fashion to give

$$\begin{aligned} &X_1X_2 + X_2X_3 + X_3X_4 + X_4X_5 \\ &+X_5X_6 + X_6X_7 + X_7X_1 + 5 \geq 0, \end{aligned} \quad (9)$$

starting from

$$\begin{aligned} &(X_1 + X_2 + X_3)^2 + (X_3 + X_4 + X_5)^2 \\ &+(X_5 + X_6 + X_7)^2 + (X_1 - X_3 + X_5)^2 \\ &+(X_1 - X_5 + X_7)^2 + 5 \geq 0 \end{aligned} \quad (10)$$

and applying the RS method. Similarly for an odd n-polygon, we obtain chained inequalities like

$$\begin{aligned} &X_1X_2 + X_2X_3 + X_3X_4 + X_4X_5 \\ &+ \cdots + X_{n-1}X_n + X_nX_1 + n - 2 \geq 0. \end{aligned} \quad (11)$$

For example, starting from

$$(X_1 - X_2 + X_3)^2 + (X_3 - X_4 + X_5)^2 + (X_1 - X_3 + X_5)^2 \geq 3 \quad (12)$$

one obtains via the RS method, the inequality $X_1X_2 + X_2X_3 + X_3X_4 + X_4X_5 - X_5X_1 \geq 3$. This kind of chained inequality [34] can be readily generalized for an odd n-polygon, to obtain inequalities of the type $X_1X_2 + X_2X_3 + X_3X_4 + X_4X_5 + \cdots + X_{n-1}X_n - X_nX_1 \geq n - 2$. To obtain LG-type inequalities via the RS procedure, we drop all particle labels. For example, writing

$$(L - K - M)^2 + (J - K + M)^2 \geq 2, \quad (13)$$

we obtain via the RS method the LG-type inequality: $\langle JK\rangle + \langle KL\rangle + \langle LM\rangle - \langle JM\rangle \leq 2$, which can be experimentally tested on qubits by measuring the properties J, K or L at time t_1, and K, L or M at time $t_2 > t_1$. However, because $\langle KL\rangle = \langle LK\rangle$ by JR, the above inequality is equivalent to the usual LG inequality.

Hybrid inequalities. – If we retain the particle labels of the original RS procedure, but do not require the *self-correlation* terms to be elimintated, the result is *hybrid* inequalities that encompass correlations paired without restriction. Self-correlation terms $\langle X_jX_k\rangle$ or $\langle Y_jY_k\rangle$ correspond to incompatible, sequential measurements, assumed to satisfy NIR, while cross-correlation terms $\langle X_jY_k\rangle$ correspond to compatible measurements assumed to satisfy LR. The resulting hybrid spatio-temporal inequality is thus a joint test of LR + NIR, *i.e.*, local noninvasive realism (LNIR).

As an example, starting from

$$(X_2 - X_1 + Y_1)^2 + (X_1 - Y_2 + Y_1)^2 \geq 2 \quad (14)$$

we obtain, via the RS method,

$$F \equiv \langle X_1X_2\rangle + \langle X_1Y_2\rangle - \langle X_2Y_1\rangle + \langle Y_1Y_2\rangle \leq 2. \quad (15)$$

The experimental protocol can be as follows: at t_1, Alice measures X_1 or not, and Bob measures Y_1 or not; at t_2, Alice measures X_2 or not, and Bob measures Y_2 or not. This gives rise to the following 16 possibilities:

$$\{\emptyset_X, X_1, X_2, X_1X_2\} \times \{\emptyset_Y, Y_1, Y_2, Y_1Y_2\}$$

where $\emptyset_X$ ($\emptyset_Y$) denotes a nonaction by A (B) while X_1X_2 or Y_1Y_2 is a pair of sequential measurements by a party on the respective particle. Of these 16, only 9 turn out to be experimentally admissible, while 3 yield two data points. The observational set corresponding to the pair $(\emptyset_X, Y_1Y_2)$ yields the data $\langle Y_1Y_2\rangle$, as does $(X_1X_2, \emptyset_Y)$ yield $\langle X_1X_2\rangle$. The pairs (X_1, Y_2) and (X_2, Y_1) yield, respectively, the data $\langle X_1Y_2\rangle$ and $\langle X_2Y_1\rangle$. The pair (X_1, Y_1Y_2) yields the data $\langle X_1Y_1\rangle$ (which does not appear in ineq. (15)) and $\langle Y_1Y_2\rangle$ but not $\langle X_1Y_2\rangle$ since measurement of Y_1 is invasive for Y_2. Similarly, the pair (X_1X_2, Y_1) contributes the data $\langle X_1X_2\rangle$. On the other hand, (X_2, Y_1Y_2) yields, apart from $\langle X_2Y_1\rangle$ also $\langle Y_1Y_2\rangle$. Similarly, the pair (X_1X_2, Y_2) contributes data $\langle X_1X_2\rangle$ and $\langle X_1Y_2\rangle$, while (X_1X_2, Y_1Y_2) yields $\langle X_1X_2\rangle$ and $\langle Y_1Y_2\rangle$.

For the Bell state $|\Psi^-\rangle = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$, we find that $F = \hat{x}_1\cdot\hat{x}_2 + \hat{x}_1\cdot\hat{y}_2 - \hat{x}_2\cdot\hat{y}_1 + \hat{y}_1\cdot\hat{y}_2$ which attains a value of $2\sqrt{2}$ for the settings where $\hat{x}_2, \hat{x}_1, \hat{y}_2, \hat{y}_1$ differ sequentially by $\pi/4$. This is in fact the Tsirelson bound [35] for this inequality. By the unification, ineq. (15) is indifferent to whether the correlata belong to the same particle or two different particles. Thus, we can re-interpret X_1, Y_1 as belonging to one particle (say A) and X_2, Y_2 as belonging to another particle (say B). As a result, the self-correlation terms in ineq. (15) become cross-correlations, while the cross-correlation terms remain so, in effect converting the spatio-temporal inequality into the CHSH inequality [1], where all correlata are cross-correlations. Now we make use of the fact that these cross-correlation terms, when evaluated for a singlet, are numerically the same as the (state-independent) temporal self-correlation term [36]. Thus, the Tsirelson bound for ineq. (15) is the same as that for the CHSH inequality, which is $2\sqrt{2}$, attained for singlets. In general, for arbitrary spin, the Tsirelson bound may be different under the conversion [25].

An interesting fact here is that product states can violate the hybrid inequality (15). This is because the hybrid inequality can be violated from contributions both from the local sector (via quantum invasiveness) and nonlocal sector, and the former, being state-independent, can be made sufficiently high for the violation even with separable states. Let the two particles be in the product state $|\psi_A\rangle|\psi_B\rangle$, where $|\psi_A\rangle\langle\psi_A| = \frac{1}{2}(1 + \hat{n}_A\cdot\vec{\sigma})$ and $|\psi_B\rangle\langle\psi_B| = \frac{1}{2}(1 + \hat{n}_B\cdot\vec{\sigma})$. Then we find for this state that

$$F = \hat{x}_1\cdot\hat{x}_2 + (\hat{x}_1\cdot\hat{n}_A)(\hat{y}_2\cdot\hat{n}_B) - (\hat{x}_2\cdot\hat{n}_A)(\hat{y}_1\cdot\hat{n}_B) + \hat{y}_1\cdot\hat{y}_2. \quad (16)$$

$$\begin{aligned}\left|\left|\sqrt{\hat{F}^2}\right|\right| &\leq \left|\left|(\hat{x}_1\cdot\hat{x}_2)+(\hat{y}_1\cdot\hat{y}_2)+\sqrt{2}\sqrt{\langle(\mathbb{I}-(\hat{x}_1\cdot\hat{x}_2)(\hat{y}_1\cdot\hat{y}_2)-((\hat{x}_1\times\hat{x}_2)\cdot\vec{\sigma})\otimes((\hat{y}_1\times\hat{y}_2)\cdot\vec{\sigma})\rangle}\right|\right| \\ &= \left|\cos\theta_1+\cos\theta_2+\sqrt{2}\sqrt{1-(\cos(\theta_1)\cos(\theta_2)+\sin(\theta_1)\sin(\theta_2))}\right| \\ &= \left|\cos\theta_1+\cos\theta_2+\sqrt{2}\sqrt{1-\cos(\theta_1-\theta_2)}\right| \leq 2\sqrt{2}, \end{aligned} \tag{17}$$

For the settings where $\hat{x}_2, \hat{x}_1, \hat{y}_2, \hat{y}_1$ differ sequentially by $\pi/4$, and $\hat{n}_A = \hat{n}_B = \hat{y}_2$, we find that ineq. (15) yields $\frac{3}{\sqrt{2}} \approx 2.12$.

The lhs in ineq. (15) is the expectation value of the correlation operator $\hat{F} = [\mathbf{X}_1\cdot\mathbf{X}_2+\mathbf{Y}_1\cdot\mathbf{Y}_2]+[(\mathbf{X}_1\cdot\vec{\sigma})\otimes(\mathbf{Y}_2\cdot\vec{\sigma})-(\mathbf{X}_2\cdot\vec{\sigma})\otimes(\mathbf{Y}_1\cdot\vec{\sigma})]$ where the first term in the square brackets (denoted S_1) is a state-independent one and the second term (denoted S_2) is the state-dependent one. Interestingly, we find after some straightforward manipulation, $(S_2)^2 = 2[\mathbb{I}-(\mathbf{X}_1\cdot\mathbf{X}_2)(\mathbf{Y}_1\cdot\mathbf{Y}_2)-((\mathbf{X}_1\times\mathbf{X}_2)\cdot\vec{\sigma})\otimes((\mathbf{Y}_1\times\mathbf{Y}_2)\cdot\vec{\sigma})]$, which is also state-independent. Putting together contributions from S_1 and S_2, we find for this state-independent quantity a Tsirelson-like behavior in that

see eq. (17) *above*

which is the same as the Tsirelson bound for the spatio-temporal ineq. (15). This is saturated for example with $\theta_1 = -\theta_2 = \pi/4$.

Monogamy inequalities. – Here we point out that the RS formalism can be used to derive an interesting monogamous behaviour between the violations of the KCBS and CHSH inequalities, which was recently obtained by Kurzynski *et al.* [37]. Such a behavior generalizes the monogamy for nonsignaling spatial correlations and nondisturbing contextuality correlations, and is suggested by the above idea of hybrid correlations.

Consider the RS requirement $\{(X_3+Y_1+Y_2)^2+(X_1+Y_1-Y_2)^2\}+\{(X_1+X_2+X_3)^2+(X_3+X_4+X_5)^2+(X_1-X_3+X_5)^2\} \geq 5$, which yields the monogamy relation as follows

$$\begin{aligned}&\langle X_3Y_1+X_3Y_2+X_1Y_1-X_1Y_2\rangle+\langle X_1X_2+ \\ &X_3X_2+X_3X_4+X_4X_5+X_5X_1\rangle \geq -5 \\ &\langle\mathrm{CHSH}\rangle_{XY}+\langle\mathrm{KCBS}\rangle_X \geq -5. \end{aligned} \tag{18}$$

Suppose that the particle X is at least 3-dimensional, and $[X_j, X_k] = 0$ iff $|j-k| \leq 1 \mod 5$. To see that eq. (18) is a monogamy relation, *i.e.*, one whose violation leads to signaling and would thus be disallowed, we re-arrange the terms in the lhs to obtain $\langle X_3Y_2-X_1Y_2+X_1X_2+X_3X_2\rangle+\langle X_3Y_1+Y_1X_1+X_5X_1+X_4X_5+X_3Y_4\rangle$, the former which has a CHSH-like form, and the latter a KCBS-like form. Assuming that no-signaling holds between particles X and Y, we can write down a JD p_C for both the CHSH-like and KCBS-like terms, such that experimentally accessible correlations p_E are recovered. For example, for the CHSH-like term, in the method of ref. [37], we can guarantee the satisfaction of the inequality by constructing a JD $p_C(x_1,x_3;y_1,x_2) = (p_E(x_1,x_2,y_2)p_E(x_2,x_3,y_2))/p_E(x_2,y_2)$, provided $\sum_{x_1} p_E(x_1,x_2,y_2) = \sum_{x_3} p_E(x_2,x_3,y_2) = p_E(x_2,y_2)$, which is the no-signaling, or generally, the no-disturbance condition.

Conclusion and discussions. – We have shown that the experimental violation of the spatial, contextual and temporal correlation inequalities are different instantiations of the same underlying property of joint realism of all observables in the experiment. As one manifestation of this unification, the classical bound is indifferent to the particular instantiation. As another manifestation, a new class of hybrid spatio-temporal inequalities has been proposed, which is shown to be maximally violated even by product states.

The unification of the correlations provides a framework to understand the relation between the security of the three broadly different quantum cryptography protocols [38]: those based on nonlocality, such as the Ekert protocol [39], those based on conjugate coding and invasive measurement, such as BB84 [40], and those based on a single orthogonal basis like the Goldenberg-Vaidman (GV) protocol [41,42] or the counterfactual quantum protocols [43,44]. The connection between JD and security is that (nonsignaling) correlations that violate a Bell-type inequality possess cryptographically desirable properties like monogamy, no-cloning, uncertainty, incompatibility and secrecy [45]. The connection to JD is manifest in the Ekert protocol, where security requires the violation of a Bell-type inequality. The correlations that arise in the BB84 does not lead to violation of LG inequality, even though it is based on incompatibility or invasiveness of local measurements. It is this lack of violation that makes BB84 insecure in the device-independent cryptography scenario [46]. Likewise in the GV protocol, though the observed data does not lead to data for which a correlation inequality can be tested, the encoding basis is incompatible with any basis accessible to an eavesdropper.

* * *

DH and SA acknowledge support from DST for Project No. SR/S2/PU- 16/2007 and the INSPIRE fellowship IF120025, respectively. DH and SA acknowledge the Centre for Science, Kolkata and Manipal University,

respectively, for research support. The authors thank the Referee for valuable suggestions.

REFERENCES

[1] Clauser J. F., Horne M. A., Shimony A. and Holt R.A., *Phys. Rev. Lett.*, **23** (1969) 880.
[2] Bell J. S., *Rev. Mod. Phys.*, **38** (1966) 447.
[3] Einstein A., Podolsky B. and Rosen N., *Phys. Rev.*, **47** (1935) 777.
[4] Kochen S. and Specker E., *J. Math. Mech.*, **17** (1967) 59.
[5] Klyachko A. A., Can M. A., Binicioğlu S. and Shumovsky A. S., *Phys. Rev. Lett.*, **101** (2008) 020403.
[6] Cabello A., Estebaranz J. and García-Alcaine G., *Phys. Lett. A*, **212** (1996) 183.
[7] Yu S. and Oh C. H., *Phys. Rev. Lett.*, **108** (2012) 030402.
[8] Leggett A. J. and Garg A., *Phys. Rev. Lett.*, **54** (1985) 857.
[9] Lapiedra R., *Europhys. Lett.*, **75** (2006) 202.
[10] Kofler J. and Brukner C., *A condition for macroscopic realism beyond the Leggett-Garg inequalities*, arXiv:quant-ph/1207.3666.
[11] Devi A. R. U., Karthik H. S., Sudha and Rajagopal A. K., *Phys. Rev. A*, **87** (2013) 052103.
[12] Emary C., Lambert N. and Nori F., arXiv:1304.5133 (2013).
[13] Aspect A., Grangier P. and Roger G., *Phys. Rev. Lett.*, **47** (1981) 460.
[14] Aspect A., Dalibard J. and Roger G., *Phys. Rev. Lett.*, **49** (1982) 1804.
[15] Weihs G., Jennewein T., Simon C., Weinfurter H. and Zeilinger A., *Phys. Rev. Lett.*, **81** (1998) 5039.
[16] Scheidl T., Ursin R., Kofler J., Ramelow S., Ma X. S., Herbst T., Ratschbacher L., Fedrizzi A., Langford N. K., Jennewein T. and Zeilinger A., *Proc. Natl. Acad. Sci. U.S.A.*, **107** (2010) 19708.
[17] Lapkiewicz R., Li P., Schaeff C., Langford N. K., Ramelow S., Wieśniak M. and Zeilinger A., *Nature*, **474** (2011) 490.
[18] Ahrens J., Amselem E., Cabello A. and Bourennane M., arXiv:1301.2887 (2013).
[19] Friedman J. R., Patel V., Chen W., Tolpygo S. K. and Lukens J. E., *Nature*, **406** (2002) 43.
[20] van der Wal C. H., ter Haar A. C. J., Wilhelm F. K., Schouten R. N., Harmans C. J. P. M., Orlando T. P., Lloyd S. and Mooij J. E., *Science*, **290** (2000) 773.
[21] Goggin M. E., Almeida M. P., Barbieri M., Lanyon B. P., O'Brien J. L., White A. G. and Pryde G. J., *Proc. Natl. Acad. Sci. U.S.A.*, **108** (2010) 1256.
[22] Fedrizzi A., Almeida M. P., Broome M. A., White A. G. and Barbieri M., *Phys. Rev. Lett.*, **106** (2011) 200402.
[23] Athalye V., Roy S. S. and Mahesh T. S., *Phys. Rev. Lett.*, **107** (2011) 130402.
[24] Katiyar H., Shukla A., Rao K. R. K. and Mahesh T. S., *Phys. Rev. A*, **87** (2013) 052102.
[25] Markiewicz M., Kurzynski P., Thompson J., Lee S.-Y., Soeda A., Paterek T. and Kaszlikowski D., arXiv:1302.3502 (2013).
[26] Roy S. M. and Singh V., *J. Phys. A: Math. Gen.*, **11** (1978) L167.
[27] Roy S. M. and Singh V., *J. Phys. A: Math. Gen.*, **12** (1979) 1003.
[28] Fine A., *Phys. Rev. Lett.*, **48** (1982) 291.
[29] Seevinck M. and Uffink J., *Phys. Rev. A*, **76** (2007) 042105.
[30] Banik M., Gazi M. R., Ghosh S. and Kar G., *Phys. Rev. A*, **87** (2013) 052125.
[31] Oppenheim J. and Wehner S., *Science*, **330** (2010) 1072.
[32] Aravinda S. and Srikanth R., arXiv:1211.6407 (2012).
[33] Barrett J., *Phys. Rev. A*, **75** (2007) 032304.
[34] Braunstein S. L. and Caves C. M., *Ann. Phys. (N.Y.)*, **202** (1990) 22.
[35] Tsirelson B. S., *Lett. Math. Phys.*, **4** (1980) 93.
[36] Fritz T., *New J. Phys.*, **12** (2010) 083055.
[37] Kurzynski P., Cabello A. and Kaszlikowski D., *Fundamental monogamy relation between contextuality and nonlocality*, arXiv:1307.6710 (2013).
[38] Shukla C., Pathak A. and Srikanth R., *Int. J. Quantum Inf.*, **10** (2012).
[39] Ekert A. K., *Phys. Rev. Lett.*, **67** (1991) 661.
[40] Bennett C. H. and Brassard G., *Quantum cryptography: Public key distribution and coin tossing*, in *Proccedings of the IEEE International Confrence on Computers, Systems, and Signal Processing, Bangalore, 1984* (IEEE) 1984, p. 175.
[41] Goldenberg L. and Vaidman L., *Phys. Rev. Lett.*, **75** (1995) 1239.
[42] Yadav P., Srikanth R. and Pathak A., *Generalization of the Goldenberg-Vaidman QKD Protocol*, arXiv:1209.4304.
[43] Noh T.-G., *Phys. Rev. Lett.*, **103** (2009) 230501.
[44] Shenoy H. A., Srikanth R. and Srinivas T., *EPL*, **103** (2013) 60008.
[45] Masanes L., Acin A. and Gisin N., *Phys. Rev. A*, **73** (2006) 012112.
[46] Acín A., Brunner N., Gisin N., Massar S., Pironio S. and Scarani V., *Phys. Rev. Lett.*, **98** (2007) 230501.